

CHUANYU PAN

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EDUCATION

University of California, Berkeley

Master of Engineering, Visual Computing and Computer Graphics in Computer Science.

Aug 2022 – May 2023

GPA: 4.0/4.0

Tsinghua University

Bachelor of Engineering, Computer Science and Technology.

Aug 2017 – Jul 2022

GPA: 3.8/4.0

Direction Highlights: Generative AI, Computer Graphics, 3D Computer Vision, Machine Learning, VR/AR, Distributed System

SKILLS

- **Programming Language:** Python(strong), C/C++(strong), CUDA, JavaScript. Proficient **Coding Agent** User.
- **Framework & Tool:** Pytorch, Profiler, Ray, Unity3D, OpenGL, OpenCV, Blender, Vulkan, React, Git.
- **Research Expertise:** Diffusion, Transformer, VAE & Representation (3D, Image, Video), Gaussian Splatting/NeRF
- **Engineering Expertise:** production-level data infrastructure, hundred-gpus parallel training, MFU & VRAM optimization.

EXPERIENCE

Meshy AI | **Foundation Model Tech Lead** (promoted)

Sunnyvale, CA | May 2025 – Now

- Lead **3D generative research** of Meshy's latest text/image to 3d model: **Meshy-5** to **Meshy-7**, from training to data.
- Train world-first **3-stage ultra-high resolution** (2048³) 3D generation model with **30B DiT** and x16 compression **SparseConv VAE**. Optimize Training efficiency with **compiled kernel** and **parallel training** (DP, FSDP, SP)
- Apply comprehensive **cleaning, labeling, analyzing, and editing** on Meshy's **10-million** level 3D dataset with a well-designed pipeline, which leads Meshy's model to have the best **prompt alignment** and **structure correctness**.
- Develop high throughput **training data ingestion pipeline** and efficient **dataloader** to unblock data loading in training.
- Lead research on Meshy 3D studio, including **polygon generation, part segmentation, watertight geometry**, etc.
- Strong technical leadership in **setting research goals, project planning, hiring, and performance review**.

Meshy AI | **3D AI Research Engineer**

Sunnyvale, CA | Nov 2023 – May 2025

- Serve as **founding engineer** for **Meshy-1** to **Meshy-4**, proposing world's **best 3D diffusion model** at the time released.
- Track and apply SOTA **diffusion models** and **3D representations (NeRF, GS)** for cutting-edge 3D generative modeling.
- Propose world-first **3D native VAE** encoding high-fidelity geometry, and **10B+ MMDiT** conditioned by text and image.
- Optimize nerf distillation process from **1hr to less than 5mins**, with better **cuda** implementation and smarter sampling.
- Design Meshy's **data ETL** and build data pipelines with Ray and Databricks, enabling **millions of** data processing in a day.
- Build Meshy's **training infra** from scratch, enabling training on **hundreds of gpus** in parallel.
- Field: Diffusion Model, Transformer, NeRF, Gaussian Splatting, Computer Graphics | Key tools: Pytorch, C++, CUDA, Ray

Honda Research Institute (HRI) | **Research Scientist Intern – Neural Rendering**

San Jose, CA | Jun 2023 – Nov 2023

- Apply **neural rendering** to LiDAR **point cloud** for 3D street scene reconstruction and forecasting.
- Build the system with **CUDA** for real-time performance and reach **SOTA accuracy** on 3D reconstruction tasks.
- Advisor: Aolin Xu | Field: 3D Vision, NeRF, LiDAR point cloud | Key tools: Pytorch, OpenCV, CUDA

Huiye Technology Inc. | **Software Engineer Intern - Animation**

Beijing | Jun 2020 - Sep 2020

- Implemented a system to drive **3D real-time facial animation** by audio input with **Pytorch** and **Unity3d**.
- Applied algorithms to a 3D avatar 'Xiaoyu,' **live streamed** on Bilibili (Youtube in China) and **attracted 312k viewers**.
- Advisor: Buyu Li | Field: Facial Animation, Multi-modal | Key tools: Pytorch, Unity3d

HIGHLIGHTED WORKS

Digital Twin Tracking Dataset (DTTD), FHL Vive Center for Enhanced Reality, UC Berkeley

Aug 2022 – Jul 2023

- Build **Digital Twin Tracking Dataset (DTTD)** with cutting edge LiDAR and motion capture sensors.
- Developed real-time **mm-error** 3D object tracking and pose estimation that aim for **general AR applications**.
- Advisor: Allen Y. Yang | Publication: **CVPR2023** | Field: 3D Object Tracking, SFM, Camera | Key tools: Pytorch, Kinect

3D Cartoon Face Reconstruction and Creation, Jittor Group, Tsinghua

Dec 2021 - Jul 2022

- Lead research on **3D cartoon face reconstruction** with **semi-supervised learning** and **mesh deformation**.
- Transformed the research to an **automatic avatar creation** API for VR online meetings with **Pytorch** and **WebGL**.
- Advisor: Shimin Hu | Publication: **VRIH** | Field: 3D Reconstruction, VR, Digital Human | Key tools: Pytorch, WebGL

Robust 3D Human Reconstruction in Seconds, 3D Vision and Computational Photography Lab

Aug 2018 - Nov 2019

- Implemented a robust system to **reconstruct 3D human bodies** with fancy clothes in **3-8 seconds** using **Azure Kinect**.
- Achieved **SOTA performance** on capturing **accurate and detailed** human shapes with **complex cloth structures**.
- Advisor: Yebin Liu | Publication: **CVPR2020 (oral)** | Field: 3D Reconstruction, Digital Human | Key tools: C++, CUDA

PUBLICATIONS (* INDICATES THE FIRST AUTHOR)

- [1] **Chuanyu Pan***, Guowei Yang, Taijiang Mu, Yukun Lai. "Generating Animatable 3D Cartoon Faces from Single Portraits" in the *Computer Graphics International (CGI)*, 2023 and *Virtual Reality & Intelligent Hardware (VRIH)* Journal.
- [2] Weiyu Feng*, Seth Z. Zhao*, **Chuanyu Pan***, Adam Chang, Yichen Chen, Zekun Wang, Allen Y. Yang. "Digital Twin Tracking Dataset (DTTD): A New RGB+Depth 3D Dataset for Longer-Range Object Tracking Applications" in the *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, the 2nd Workshop Challenge on **Vision Datasets Understanding**, 2023
- [3] **Chuanyu Pan***, Yanchao Yang*, Kaichun Mo, Yueqi Duan, and Leonidas J. Guibas. "Object Pursuit: Building a Space of Objects via Discriminative Weight Generation" in *ICLR*, 2022
- [4] Zhe Li, Tao Yu, **Chuanyu Pan**, Zerong Zheng, and Yebin Liu. "Robust 3D Self-portraits in Seconds" in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020 (Oral Presentation)
- [5] Huang Zixun*, Keling Yao*, Seth Z. Zhao*, **Chuanyu Pan***, Tianjian Xu, Weiyu Feng, and Allen Y. Yang. "Towards Subcentimeter Accuracy Digital-Twin Tracking via An RGBD-based Transformer Model and A Comprehensive Mobile Dataset." in the *International Conference on Machine Learning (ICML)* 2024.
- [6] **Chuanyu Pan***, Aolin Xu. "LiDARGrid: 3D Opacity Grid from LiDAR for Scene Forecasting" in the *International Conference on Robotics and Automation (ICRA)* 2024.
- [7] Yihao Luo, Xianglong He, **Chuanyu Pan**, Yiwen Chen, Jiaqi Wu, Yangguang Li, Wanli Ouyang, Yuanming Hu, Guang Yang and ChoonHwai Yap. "Faithful Contouring: Near-Lossless 3D Voxel Representation Free from Iso-surface" in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 (Oral)
- [8] Chongjie Ye, Cheng Cao, **Chuanyu Pan**, Yiming Hao, Yihao Zhi, Yuanming Hu, and Xiaoguang Han. "Omni123: Exploring 3D Native Foundation Models with Limited 3D Data by Unifying Text to 2D and 3D Generation." *arXiv preprint arXiv:2604.02289* (2026).
- [9] Kim Jeonghwan, Yushi Lan, Yongwei Chen, Hieu Trung Nguyen, **Chuanyu Pan**, and Xingang Pan. "SceneConductor: 3D Scene Generation from a Single Image with Multi-Agent Orchestration." (2026)
- [10] Li Zhengyuan, **Chuanyu Pan**, Yuanming Hu, Raymond Yeh. "Branch-Centric Tokenization and Test-Time Augmentation for Skeleton Generation." *arXiv preprint arXiv:2609.06218* (2026).
- [11] Chang, Jiahao, Dong Du, Wanhui Sun, Yujian Zheng, **Chuanyu Pan**, Bowen Zhao, Chongjie Ye, Yuanming Hu, and Xiaoguang Han. "SAM3D-Part: Interactive Part Selection and Generation from 3D Objects." *ACM SIGGRAPH Asia 2026 Conference Papers*